LAB 5 USER INTERFACE DESIGN, FIRST WORKING PROTOTYPE, AND DELIVERABLE I PREPARATION

COURSE: SYSTEM DEVELOPMENT COURSE CODE: 420-940-VA WEEK: 5 LAB DURATION: 5 HOURS

INTRODUCTION

By Week 5 the project must move beyond paper based planning. Previous weeks establish the foundation through requirements, user stories, and UML diagrams. Week 5 requires the project to become visible and demonstrable through screens and early working behavior. The course outline lists User Interface Design, Project Presentation, Deliverable I, and the Deliverable I Report for this week. This lab supports those exact outcomes.

MAIN FOCUS OF THIS LAB

- Conversion of user stories into visible screens
- Design of a simple but logical interface
- Construction of a first working C# Windows Forms prototype
- Addition of basic event-driven behavior
- Validation of user input
- Preparation of evidence for Deliverable I

Lab Objectives Participants achieve the following:

- Design of at least two interface screens for the project
- Selection of appropriate controls such as labels, text boxes, buttons, and message boxes
- Construction of a simple Windows Forms prototype in C#
- Writing of basic event-driven code for button clicks
- Addition of simple input validation and user feedback
- Connection of interface elements to earlier requirements and user stories
- Preparation of screenshots, explanation notes, and working material for Deliverable I

Required Materials

- Access to Omnivox
- Project work from Weeks 1, 2, 3, and 4
- Requirements, user stories, and UML diagrams
- Visual Studio with the .NET desktop development tools installed
- Word processor (Microsoft Word or Google Docs)
- Screenshot tool

Suggested Submission Package One compressed file is uploaded to Omnivox: **Lab5_UI_Prototype_LastName.zip**

The compressed file contains:

1. The complete Visual Studio project folder
2. The lab document named **Lab5_UI_Prototype_LastName.docx**

The lab document contains:

- Project summary
- Wireframes or interface sketches
- Screenshots of working forms
- Short explanations for each screen
- Reflection answers

Lab Structure and Timing The lab is structured as a full 5-hour session.

PHASE 1: REVIEW OF PROJECT AND SELECTION OF FIRST PROTOTYPE SCREENS TIME: 30 MINUTES

Previous project work is opened:

- Lab 1 project idea
- Lab 2 requirement document
- Lab 3 user stories and iteration plan
- Week 4 UML diagrams

A section titled **1. Project Summary for Week 5** is created in the lab document. One paragraph states:

- The project name
- The main users
- The problem solved
- The user stories selected for Deliverable I

Two screens to be built in the lab are identified. Examples: Appointment Request Form, Advisor Dashboard, Login Form, Ticket Submission Form, Inventory Entry Form, Event Registration Form.

Purpose of this phase Interface work remains connected to earlier analysis.

PHASE 2: CREATION OF FIRST WIREFRAMES TIME: 45 MINUTES

A section titled **2. Wireframes and Interface Plan** is created in the lab document. A rough wireframe is produced for each of the two selected screens using draw.io, Word shapes, PowerPoint shapes, or a photographed hand-drawn sketch.

Each wireframe shows:

- Screen title
- Labels
- Input fields
- Buttons
- Main navigation actions

A short explanation appears under each wireframe:

- Purpose of the screen
- Supported user story
- Reason for selected controls

Purpose of this phase Design remains centered on the user task before coding begins.

PHASE 3: CREATION OF THE WINDOWS FORMS PROJECT AND SCREEN LAYOUTS TIME: 60 MINUTES

Visual Studio is opened and a new Windows Forms App project is created in C#. The project receives a meaningful name such as AdvisingPrototype or TicketSystemPrototype.

The two planned screens are built. The first form includes at minimum: one title label, at least two input controls, and at least two buttons. The second form includes at minimum: one title label, one display or action area, and one navigation button (Back, Close, or View Details).

Controls are renamed clearly (examples: lblTitle, txtStudentNumber, txtReason, btnSubmit, btnClear, btnViewRequests). Default names are avoided where possible.

A section titled **3. Screen Layout Decisions** is created in the lab document. Four to six sentences explain:

- Organization of controls
- Logical layout
- Support for clarity and usability

Purpose of this phase The project moves from wireframe to actual interface layout.

PHASE 4: ADDITION OF EVENT-DRIVEN BEHAVIOR IN C# TIME: 75 MINUTES

The interface is made responsive. The first form includes: Submit button, Clear button, at least one validation rule, at least one feedback message using a message box, and one button that opens the second form.

A section titled **4. C# Event-Driven Behavior** is created in the lab document. Key code is copied into the document with explanations.

The following sample structure is adapted (control names and messages are changed to match the project):

Example 1: Submit button with validation

C#

```
private void btnSubmit_Click(object sender, EventArgs e) // This method runs when the user clicks the Submit button.

{

    string studentNumber = txtStudentNumber.Text; // Reads the text from the student number field and stores it in a variable.


    string reason = txtReason.Text; // Reads the text from the reason field and stores it in a variable.


    if (studentNumber == "" || reason == "") // Checks whether either required field is empty.

    {
```

```
        MessageBox.Show("Please complete all required fields before submitting.", "Missing Information"); // Shows an error message to the user.

        return; // Stops the method so the success message does not appear when the form is incomplete.

    }


    MessageBox.Show("Request submitted successfully.", "Success"); // Shows a success message if validation passed.

}
```

Example 2: Clear button

C#

```
private void btnClear_Click(object sender, EventArgs e) // This method runs when the user clicks the Clear button.

{

    txtStudentNumber.Text = ""; // Clears the student number field.

    txtReason.Text = ""; // Clears the reason field.

    txtStudentNumber.Focus(); // Moves the cursor back to the first field.

}
```

Example 3: Open second form

C#

```
private void btnOpenDashboard_Click(object sender, EventArgs e) // This method runs when the user clicks the button that opens the second form.

{

    Form2 dashboardForm = new Form2(); // Creates a new object from the second form.

    dashboardForm.Show(); // Displays the second form on the screen.

}
```

The code is tested with good input, empty input, Clear button, and the second form. In the lab document each button, validation rule, and user feedback is explained in plain language.

Purpose of this phase The interface becomes functional through event-driven C# behavior.

PHASE 5: IMPROVEMENT OF THE INTERFACE THROUGH TESTING AND SMALL FIXES TIME: 45 MINUTES

The prototype is tested as a user would use it. The following items are checked: label clarity, obvious required fields, correct button operation, understandable error messages, easy-to-follow layout, and working navigation.

Screenshots are taken of the first form, second form, one validation message, and one success message. These screenshots are pasted into a section titled **5. Testing and Interface Improvements**.

Answers are provided for:

- Observations made during testing
- Small improvements performed after testing
- Items still requiring improvement later

Purpose of this phase Interface design is linked to refinement as required in Agile work.

PHASE 6: PREPARATION OF DELIVERABLE I SUPPORT MATERIAL TIME: 45 MINUTES

A final section titled **6. Deliverable I Preparation Notes** is created. Short paragraph answers address:

1. User stories already visible in the prototype
2. Elements that currently work in the interface
3. Parts that remain only partially complete
4. Improvements planned before or during Deliverable I
5. First item to be demonstrated if the prototype is presented in class

A short list titled **What can be shown in Deliverable I** is added. Items include: working first form, working second form, validation message, clear button behavior, navigation between forms, and screenshots for the report.

Purpose of this phase The lab connects directly to the official Week 5 project expectations (Project Presentation, Deliverable I, and Deliverable I Report).

What Must Be Submitted One compressed file: **Lab5_UI_Prototype_LastName.zip**

Contents:

- Complete Visual Studio project folder
- Lab document **Lab5_UI_Prototype_LastName.docx**

Required Order of Sections in the Lab Document

1. Project Summary for Week 5
2. Wireframes and Interface Plan
3. Screen Layout Decisions
4. C# Event-Driven Behavior
5. Testing and Interface Improvements
6. Deliverable I Preparation Notes

Sample Mini Model Example project Advising Appointment System

Example first screen Appointment Request Form

Example second screen Advisor Dashboard

Example explanation The first screen supports the user story "As a student, I want to request an advising appointment so that I can avoid email delays." The second screen supports the advisor's need to view requests. Text boxes are used because the user must type data, and buttons are used because the main actions are Submit, Clear, and Open Dashboard.

Example improvement note Testing showed that the user could click Submit without entering a reason. Validation was added to stop submission and display a clearer message box. This made the form easier to understand.

Extra sample reflection Testing also showed that the Submit button was too close to the Clear button and could be clicked by accident. Spacing was added between the buttons and the Submit button was given a brighter colour so it stands out as the main action.

Instructor Note This lab is built directly from the Week 5 outline items even though the outline does not name a separate Week 5 lab explicitly. It supports User Interface Design, Project Presentation, Deliverable I, and the Deliverable I Report. The lab respects the course direction that C# is used for event-driven programming, user interfaces, and later database connections, while keeping the work at a realistic first-prototype level.

COMPLETE SAMPLE STUDENT LAB DOCUMENT (MODEL ANSWERS) LAB5_UI_PROTOTYPE_SMITH.DOCX (EXAMPLE SUBMISSION)

1. Project Summary for Week 5 The project is an Advising Appointment System for Vanier College students. The main users are students and academic advisors. The system solves the problem of slow email-based appointment booking by allowing students to request appointments online and advisors to manage them easily. For Deliverable I the following user stories are shown: "As a student, I want to request an advising appointment" and "As an advisor, I want to view pending requests".

2. Wireframes and Interface Plan

Screen 1: Appointment Request Form (Imagine a simple sketch here showing title, Student Number textbox, Reason textbox, Date picker, Submit and Clear buttons.) This screen lets students submit a request. It supports the student user story. Text boxes are used for free text input and buttons for the main actions because they are clear and familiar.

Screen 2: Advisor Dashboard (Imagine a sketch showing title, list of pending requests in a grid, View Details button.) This screen shows advisors their pending appointments. It supports the advisor user story. A grid control is used for easy reading and a button for navigation.

3. **Screen Layout Decisions**:
   The title label is placed at the top centre for clarity. Input fields are stacked vertically with clear labels on the left. Buttons are placed at the bottom in a row so they are easy to reach. The layout follows a logical top-to-bottom flow that matches how people fill forms. This design supports clarity and usability by keeping related items together and using consistent spacing.
4. **C# Event-Driven Behavior**
   The code examples from the lab are used with names changed to match the project.

- The Submit button checks that both fields are filled and shows a success or error message.
- The Clear button empties the fields and returns the cursor to the first box.
- The Open Dashboard button opens the second form. The validation rule prevents empty submissions and gives the user immediate feedback.

5. **Testing and Interface Improvements**
   Testing showed that the Submit button was too close to the Clear button. Spacing was added and the Submit button was made green. The error message was clear and helpful. A real date picker still needs to replace the text box for the appointment date. Navigation between the two forms worked perfectly.

6. Deliverable I Preparation Notes

1. Both main user stories are now visible in the prototype.
2. The request form and dashboard both work and respond to button clicks.
3. The dashboard only shows dummy data for now.
4. Real data loading and a proper date control will be added before the presentation.
5. In class the student request form will be shown first, followed by submission of a request and then the advisor dashboard to show the new request.

What can be shown in Deliverable I

- Working first form (Appointment Request)
- Working second form (Advisor Dashboard)
- Validation message
- Clear button behavior
- Navigation between forms
- Screenshots in report